

Arghamitra Talukder

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EDUCATION

2023 - 2028	Ph.D. in Computer Science, Columbia University Advisor: Itzik Pe'er , David A. Knowles <i>NSF CSGrad4US Fellow</i>	New York, NY GPA: 3.98/4.0
2018 - 2021	B.S. in Electrical Engineering, Texas A&M University <i>Engineering Honors, Magna Cum Laude</i>	College Station, TX GPA: 3.78/4.0

PROJECTS

Learning Interpretable Representations via Privileged-Basis Embeddings

Designed a training framework that enforces hierarchical information allocation across embedding dimensions, yielding a privileged basis with ordered feature importance and improved interpretability of latent representations.

Predicting Exon Inclusion Levels via Contrastive Sequence Models

Developed a deep learning model to analyze gene regulation with a contrastive learning framework. This model identifies conserved functional elements by mapping evolutionarily related DNA sequences into a shared latent space, providing novel insights into gene expression.

Modeling cell plasticity with integrated single-cell data

Designed a statistical model to quantify cellular plasticity by integrating single-cell lineage and gene expression data. This approach utilizes a continuous annotation distribution methodology to statistically detect and score plastic cell states.

Bayesian hierarchical isoform quantification with hybrid sequencing

Developed JOLI, a novel hierarchical model that enhances RNA transcript quantification by jointly integrating short-read (SR) and long-read (LR) sequencing data. The model utilizes a multi-sample learning framework to improve estimation accuracy and reproducibility, particularly for low-abundance isoforms.

PUBLICATIONS

Arghamitra Talukder, Philippe Chlenski, and Itzik Pe'er (2026). "Objective-Specific Privileged Bases via Full-Prefix Matryoshka Learning". In: *ICML Workshop: Weight-Space Symmetries (Spotlight)*. [Online version](#).

Arghamitra Talukder, Nicholas Keung, et al. (2026). "CLADES — Contrastive Learning Augmented Differential Splicing with Orthologous Positive Pairs". In: *RECOMB*. [Online version](#).

Arghamitra Talukder, Shree Thavarekere, et al. (2025). "Multi-sample, multi-platform isoform quantification using empirical Bayes". In: *RECOMB-Seq (Short-talk)*. [Online version](#).

Arghamitra Talukder and Itsik Pe’er (2024). “Computational Method to Detect Cancerous Plasticity”. In: *IICD Workshop on Single-Cell Data Integration & OT, CRA-WP Grad Cohort for IDEALS (Short-talk)*.

Arghamitra Talukder, Rujie Yin, et al. (2022). “Does inter-protein contact prediction benefit from multi-modal data and auxiliary tasks?” In: *MLSB, NeurIPS*. [Online version](#).

Ibrahim, Bassem, **Arghamitra Talukder**, and Roozbeh Jafari (2020). “Multi-source multi-frequency bio-impedance measurement method for localized pulse wave monitoring”. In: *EMBC, IEEE*. [Online version](#).

WORK EXPERIENCE

Computational Engineering Intern, LLNL

June 2026 - Aug 2026

- Mentored by [Alan Kaplan](#) at *Lawrence Livermore National Laboratory (LLNL)*: Evaluated the capabilities and limitations of genomic foundation models, for modeling ALS-associated gene regulation and interactions.
- Benchmarked foundation model predictions against multimodal genomic data to characterize where learned representations capture—or fail to capture—disease-relevant biological signals.

Product & Test Engineer, Texas Instruments

June 2021 - Aug 2023

- Responsible for TI customized Ultra-High Voltage test technology with 70% cost efficiency.
- Created a product specific test program converting from legacy tester to the latest technology reducing 40% of the test time.
- Enabled traceability with the accuracy of 99.9% by device ID recovery obtained using OCR technology during internship.

FELLOWSHIPS

RECOMB Travel Fellowship

May 2026

NIH T15 Training Grant

Sep 2025 - Aug 2027

CISE CSGrad4US Fellowship, CRA, NSF

Sep 2023 – Aug 2025 (Tenure)

Sep 2027 – Aug 2028 (Tenure)

SKILLS

Programming & ML	Python, PyTorch, PyTorch Lightning, MATLAB, Perl, C, Bash
Genomics & Data	Genomic sequence processing, RNA-seq (short- and long-read)
Systems & Cloud	High-performance computing (SLURM, multi-GPU clusters), distributed training, Linux environments
Engineering & Testing	Python/Perl/C-based semiconductor test program development, legacy tester migration and test time reduction
Tools & DevOps	Git, Hydra/OmegaConf, WandB, Docker (basic)